UNIT 5 · TOPIC 9

Design of Tractor Controls — Hand and Foot

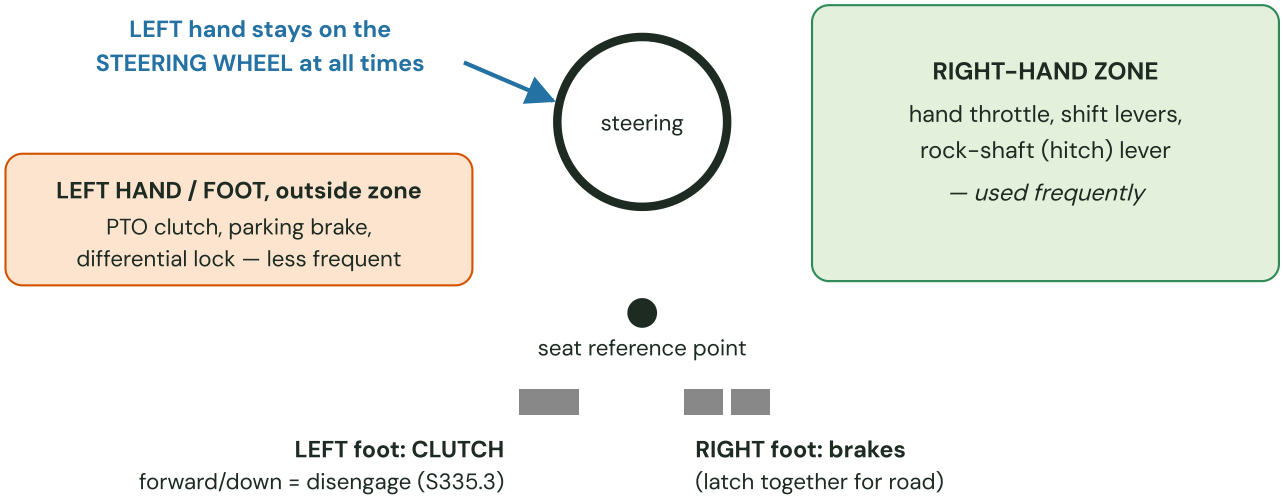
Syllabus topic: Design of tractor controls: hand and foot controls

Design philosophy of the operator workplace

Hansson et al. (quoted at Liljedahl p. 215) — design the workplace for safety, comfort and convenience:

- Good VISIBILITY from the driving position without an awkward, tiring posture.
- Levers, pedals and instruments located conveniently and logically.
- A workplace that fits both TALL AND SHORT operators.
- The operator can CHANGE WORKING POSITION easily.
- No sharp edges or obstructions (such as transmission cases) in the work area.
- **Design tools:** graphical layout plus anthropometric data are the basic tools for seat placement and control location (p. 227); Figs. 9-13/9-14 give the classical side- and top-view solution for optimum hand and foot control positions (Purcell 1980).
- Tractors are marketed worldwide, so ETHNIC (population-specific) anthropometric data must be used (p. 227) — the argument for Indian CIAE data in Indian designs.

Where controls go (p. 229)



Control location zones — plan view, after Purcell 1980 / Liljedahl Figs. 9-13/9-14 (redrawn)

Control group	Location
PRIMARY hand controls used continuously — hand throttle, shift levers, rock-shaft (hitch) lever	Inside the optimum zone, TO THE RIGHT of the operator — the LEFT hand stays free for steering at all times
Less frequent — PTO clutch, parking brake, differential lock	Left hand or foot, OUTSIDE the primary zones

- Colour coding of hand controls per **ASAE EP443** aids identification (p. 229).

Uniform location and motion — ASAE S335.3

ASAE S335.3 "Operator Controls on Agricultural Equipment" fixes uniform LOCATION and DIRECTION OF MOTION for brake, clutch, engine-speed, ground-speed (transmission), lift and steering controls, so an operator moving between makes is not surprised.

> **Clutch rule (clause 5.1.1, p. 228):** a foot clutch pedal is worked by the **LEFT foot**, moving **forward and/or downward to DISENGAGE**.

Control design principles (exam list)

- **Compatibility** — movement matches expectation (push/forward = increase/engage).
- **Consistency** across the product line.
- Adequate size and spacing to prevent inadvertent operation.
- Control–display association.
- Operating forces within strength limits (Topic 1: sustained ≤ ~30% MVC; foot pedals take higher forces than hand levers).
- **Shape coding** for blind identification (use shape + position, not colour alone).
- Guards/detents on critical controls (PTO).
- Universal SYMBOLS per **ASAE S304.5 / SAE J389b** (p. 229) — PTO, differential lock, engine rpm, fuel, coolant temperature, oil pressure, etc.

Practice MCQs

Q126. Frequently used hand controls should be located:

(a) To the operator's LEFT
(b) To the operator's RIGHT, leaving the left hand for steering
(c) Behind the seat
(d) On the floor

Q127. Which controls are assigned OUTSIDE the primary zone (left hand/foot)?

(a) Hand throttle
(b) Shift levers
(c) PTO clutch, parking brake, differential lock
(d) Rock-shaft lever

Q128. Uniform location and direction of motion of operator controls is specified in:

(a) ASAE S335.3
(b) SAE J899
(c) ISO 2631
(d) IS 9129

Q129. Per that standard, a foot clutch pedal shall be operated by the:

(a) Right foot, upward
(b) Left foot, forward and/or downward to disengage
(c) Either foot
(d) Hand

Q130. Universal symbols for operator controls are published in:

(a) ASAE S304.5 / SAE J389b
(b) SAE J185
(c) IS 4931
(d) ISO 11783

Q131. Colour coding of hand controls is recommended in:

(a) ASAE EP443
(b) SAE J209
(c) ISO 730
(d) IS 6316

Q132. Controls used blind should be distinguishable by:

(a) Colour only
(b) Shape coding (and position)
(c) Weight
(d) Sound

Q139. Because tractors are marketed worldwide, designers must use:

(a) Only US data
(b) Ethnic (population-specific) anthropometric data
(c) Averages only
(d) No data

Q140. Hansson et al. require that the operator should be able to:

(a) Stand throughout
(b) Change working position easily, with the area free of sharp edges/obstructions
(c) Work without instruments
(d) Reach all controls with feet

Answers

Q126 — (b) To the operator's RIGHT, leaving the left hand for steering — *p. 229 — right-hand primary zone; the left hand steers.*

Q127 — (c) PTO clutch, parking brake, differential lock

Q128 — (a) ASAE S335.3

Q129 — (b) Left foot, forward and/or downward to disengage — *p. 228 — clutch-pedal convention in ASAE S335.3.*

Q130 — (a) ASAE S304.5 / SAE J389b

Q131 — (a) ASAE EP443

Q132 — (b) Shape coding (and position)

Q139 — (b) Ethnic (population-specific) anthropometric data

Q140 — (b) Change working position easily, with the area free of sharp edges/obstructions

Sources: Liljedahl, Turnquist, Smith & Hoki — Tractors and their Power Units, Ch. 9 (spatial/visual/control requirements pp. 226–231; seating pp. 219–224); SAE J899 / J1163 / J898 / J185 / J154a / J209; ASAE S335.3, S304.5 / SAE J389b, EP443; Purcell (1980), Stikeleather (1981); NPTEL lectures.